

MATH 3312, SPRING 2026: HOMEWORK 6

- (1) Let V be an n -dimensional vector space, and let $T : V \rightarrow V$ be a linear transformation.
- (a) Show that any subset of V with more than n elements is linearly *dependent* (that is, it is *not* linearly independent).
 - (b) Show that, for any $v \in V$, there exist $a_0, a_1, \dots, a_n \in k$, not all zero, such that

$$a_0v + a_1T(v) + a_2T^2(v) + \dots + a_{n-1}T^{n-1}(v) + a_nT^n(v) = 0.$$

Here, T^k is simply T iterated k times.

Hint: Apply part (a) to the $n + 1$ vectors $\{v, T(v), \dots, T^n(v)\}$

- (c) For any polynomial $p(x) \in k[x]$ with $p(x) = b_mx^m + b_{m-1}x^{m-1} + \dots + b_1x + b_0$, show that the assignment

$$v \mapsto b_mT^m(v) + b_{m-1}T^{m-1}(v) + \dots + b_1T(v) + b_0v$$

is a linear transformation.

We will denote this linear transformation by $p(T)$.

- (d) If $r(x) = p(x)q(x)$, show that $r(T) = p(T) \circ q(T)$.
- (e) Show that there exists a non-zero polynomial $p(x) \in k[x]$ such that $p(T) = 0$ is the trivial linear transformation. That is,

$$p(T)(v) = 0,$$

for all $v \in V$.

Hint: Use a basis $\{v_1, \dots, v_n\}$ for V and find a polynomial such that $p(T)(v_i) = 0$ for all i .

Two $n \times n$ -matrices $A, B \in M_n(k)$ are **k -conjugate** if there exists an invertible matrix $S \in M_n(k)$ such that $A = SBS^{-1}$.

- (2) Show that the following are equivalent for $A, B \in M_n(k)$:
- (a) A and B are k -conjugate.
 - (b) There exists an n -dimensional space V with two bases $\mathcal{B}_1, \mathcal{B}_2$ and a linear transformation $T : V \rightarrow V$ such that

$$A = [T]_{\mathcal{B}_1} ; B = [T]_{\mathcal{B}_2}.$$

- (3) Show that the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is $\mathbb{C}$ -conjugate but not $\mathbb{R}$ -conjugate to a diagonal matrix.

An $n \times n$ -matrix is **upper triangular** if it doesn't have any non-zero entries below the diagonal. That is, if $a_{i,j} = 0$ whenever $i > j$.

- (4) Show that the following are equivalent for a matrix $A \in M_n(k)$ (here, T_A is the linear transformation $k^n \rightarrow k^n$ with matrix A):
- (a) A is k -conjugate to an upper triangular matrix $B = [b_{i,j}] \in M_n(k)$.
 - (b) There exists a basis $\{v_1, v_2, \dots, v_n\}$ of k^n such that, for each j ,

$$T_A(v_j) = \sum_{i \leq j} b_{i,j}v_i,$$

for some $b_{i,j} \in k$.

(c) There exists a basis $\{v_1, v_2, \dots, v_n\}$ of k^n such that, for each i ,

$$T_A(\text{span}_k(v_1, \dots, v_i)) \leq \text{span}_k(v_1, \dots, v_i).$$

A **k -pair** is a pair (V, T) , where V is a k -vector space and $T : V \rightarrow V$ is a k -linear transformation. Suppose that we have a collection of k -pairs

$$(V_1, T_1), \dots, (V_r, T_r).$$

The **product k -pair** of such a collection, denoted

$$(V_1, T_1) \times \dots \times (V_r, T_r)$$

is the k -pair with k -vector space $V_1 \times \dots \times V_r$ and linear transformation

$$(v_1, \dots, v_r) \mapsto (T_1(v_1), \dots, T_r(v_r)).$$

(5) Suppose that we have a collection as above and bases $\mathcal{B}_1, \dots, \mathcal{B}_r$ for $V_1, \dots, V_r$ such that $[T]_{\mathcal{B}_i} = A_i$. Then the product k -pair is isomorphic to (k^n, T_A) where:

- $n = \dim_k V_1 + \dots + \dim_k V_r$.
- A is the **block diagonal matrix**

$$\begin{bmatrix} A_1 & 0 & \dots & 0 \\ 0 & A_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_r \end{bmatrix}$$